

Quiz 4 — Instructor key

For Lecture 5; assesses Lecture 4

Private assessment QUIZ04, revision 1; LaTeX typography edition 1. Five one-page A4 forms; 10 minutes; 3 points. Question order: easy / medium / hard.

Give each student one form. Distribute different forms within the same sitting. The five forms assess the same three questions with different numerical values and shuffled answer choices. Keep this key separate from student printouts.

Form	Q1	Q2	Q3
4H8M	E	B	A
4HPW	G	F	C
423M	F	G	D
4GLW	A	D	B
4S6B	D	C	E

Question 1: Calculate the short population slope, distinguishing the causal effect from omitted-variable distortion.

Sources: Lecture 2 deck, omitted-variable bias; Practice Set 4, Questions 2–4.

Question 2: Complete the LLN denominator and CLT sampling-SD steps inside the OLS derivation.

Sources: Lecture 2 deck, “The LLN stabilizes the denominator” and “Combine the LLN and CLT for the slope”; Practice Set 4, Questions 1 and 8–10.

Question 3: Combine correlation, confounding, causal sign reversal, and large-sample concentration around the wrong causal target.

Sources: Lecture 2 deck, causal model, omitted-variable bias, and consistency; Practice Set 4, Questions 6, 9, and 13–15 (combined extension).

Scope: Delivered Lecture 4, from omitted-variable bias onward, combined with the earlier causal discussion. No hypothesis tests or confidence intervals. Difficulty and timing are design judgments, not student-piloted measurements.

Form 4H8M — worked solutions

Question 1 (easy). Correct answer: E, 6.

1. The causal model is

$$Y = \alpha + 7X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

The auxiliary population slope and omitted-variable distortion are

$$\pi = \frac{\text{Cov}(X, Z)}{\text{Var}(X)} = \frac{-2}{6} = -\frac{1}{3}, \quad \delta\pi = 3 \left(-\frac{1}{3} \right) = -1.$$

Hence $b_{\text{short}} = \beta + \delta\pi = 7 - 1 = 6$. The association is one unit below the causal effect. This is a population projection identity, not a formula for exact finite-sample unconditional bias.

Question 2 (medium). Correct answer: B, (4; 0.3).

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 144)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

The LLN gives $D_n \xrightarrow{p} \text{Var}(X) = 4$. Dividing the numerator by 4 divides its variance by $4^2 = 16$:

$$\sqrt{n}(\hat{b} - \beta) \xrightarrow{d} \mathcal{N}\left(0, \frac{144}{16}\right) = \mathcal{N}(0, 9).$$

Undoing the $\sqrt{n}$ scaling gives

$$\text{SD}(\hat{b}) \approx \sqrt{\frac{9}{100}} = \frac{3}{\sqrt{100}} = 0.3.$$

This is the SD of the large-sample approximation, not an exact finite-sample formula.

Question 3 (hard). Correct answer: A, (-2.5; 1).

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 0.5$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

Convert the correlation into the auxiliary regression slope:

$$\text{Cov}(X, Z) = \frac{1}{2} \cdot 2 \cdot 4 = 4, \quad \pi = \frac{4}{2^2} = 1.$$

Thus $\delta\pi = 3$ and $\beta = b_{\text{short}} - \delta\pi = 0.5 - 3 = -2.5 < 0$. But the short OLS slope converges to the positive association:

$$\hat{b}_{\text{short}} \xrightarrow{p} 0.5 > 0, \quad \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) = 1.$$

More data make the positive association increasingly precise even though the causal effect is negative. Neither a large sample nor sample residual orthogonality repairs the omitted-variable distortion.

Form 4HPW — worked solutions

Question 1 (easy). Correct answer: G, 7.

1. The causal model is

$$Y = \alpha + 8X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

The auxiliary population slope and omitted-variable distortion are

$$\pi = \frac{\text{Cov}(X, Z)}{\text{Var}(X)} = \frac{-2}{6} = -\frac{1}{3}, \quad \delta\pi = 3 \left(-\frac{1}{3} \right) = -1.$$

Hence $b_{\text{short}} = \beta + \delta\pi = 8 - 1 = 7$. The association is one unit below the causal effect. This is a population projection identity, not a formula for exact finite-sample unconditional bias.

Question 2 (medium). Correct answer: F, (4; 0.4).

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 256)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

The LLN gives $D_n \xrightarrow{p} \text{Var}(X) = 4$. Dividing the numerator by 4 divides its variance by $4^2 = 16$:

$$\sqrt{n}(\hat{b} - \beta) \xrightarrow{d} \mathcal{N}\left(0, \frac{256}{16}\right) = \mathcal{N}(0, 16).$$

Undoing the $\sqrt{n}$ scaling gives

$$\text{SD}(\hat{b}) \approx \sqrt{\frac{16}{100}} = \frac{4}{\sqrt{100}} = 0.4.$$

This is the SD of the large-sample approximation, not an exact finite-sample formula.

Question 3 (hard). Correct answer: C, (-2; 1).

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 1$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

Convert the correlation into the auxiliary regression slope:

$$\text{Cov}(X, Z) = \frac{1}{2} \cdot 2 \cdot 4 = 4, \quad \pi = \frac{4}{2^2} = 1.$$

Thus $\delta\pi = 3$ and $\beta = b_{\text{short}} - \delta\pi = 1 - 3 = -2 < 0$. But the short OLS slope converges to the positive association:

$$\hat{b}_{\text{short}} \xrightarrow{p} 1 > 0, \quad \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) = 1.$$

More data make the positive association increasingly precise even though the causal effect is negative. Neither a large sample nor sample residual orthogonality repairs the omitted-variable distortion.

Form 423M — worked solutions

Question 1 (easy). Correct answer: F, 8.

1. The causal model is

$$Y = \alpha + 9X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

The auxiliary population slope and omitted-variable distortion are

$$\pi = \frac{\text{Cov}(X, Z)}{\text{Var}(X)} = \frac{-2}{6} = -\frac{1}{3}, \quad \delta\pi = 3 \left(-\frac{1}{3} \right) = -1.$$

Hence $b_{\text{short}} = \beta + \delta\pi = 9 - 1 = 8$. The association is one unit below the causal effect. This is a population projection identity, not a formula for exact finite-sample unconditional bias.

Question 2 (medium). Correct answer: G, (4; 0.5).

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 400)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

The LLN gives $D_n \xrightarrow{p} \text{Var}(X) = 4$. Dividing the numerator by 4 divides its variance by $4^2 = 16$:

$$\sqrt{n}(\hat{b} - \beta) \xrightarrow{d} \mathcal{N}\left(0, \frac{400}{16}\right) = \mathcal{N}(0, 25).$$

Undoing the $\sqrt{n}$ scaling gives

$$\text{SD}(\hat{b}) \approx \sqrt{\frac{25}{100}} = \frac{5}{\sqrt{100}} = 0.5.$$

This is the SD of the large-sample approximation, not an exact finite-sample formula.

Question 3 (hard). Correct answer: D, (-1.5; 1).

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 1.5$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

Convert the correlation into the auxiliary regression slope:

$$\text{Cov}(X, Z) = \frac{1}{2} \cdot 2 \cdot 4 = 4, \quad \pi = \frac{4}{2^2} = 1.$$

Thus $\delta\pi = 3$ and $\beta = b_{\text{short}} - \delta\pi = 1.5 - 3 = -1.5 < 0$. But the short OLS slope converges to the positive association:

$$\hat{b}_{\text{short}} \xrightarrow{p} 1.5 > 0, \quad \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) = 1.$$

More data make the positive association increasingly precise even though the causal effect is negative. Neither a large sample nor sample residual orthogonality repairs the omitted-variable distortion.

Form 4GLW — worked solutions

Question 1 (easy). Correct answer: A, 9.

1. The causal model is

$$Y = \alpha + 10X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

The auxiliary population slope and omitted-variable distortion are

$$\pi = \frac{\text{Cov}(X, Z)}{\text{Var}(X)} = \frac{-2}{6} = -\frac{1}{3}, \quad \delta\pi = 3 \left(-\frac{1}{3} \right) = -1.$$

Hence $b_{\text{short}} = \beta + \delta\pi = 10 - 1 = 9$. The association is one unit below the causal effect. This is a population projection identity, not a formula for exact finite-sample unconditional bias.

Question 2 (medium). Correct answer: D, (4; 0.6).

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 576)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

The LLN gives $D_n \xrightarrow{p} \text{Var}(X) = 4$. Dividing the numerator by 4 divides its variance by $4^2 = 16$:

$$\sqrt{n}(\hat{b} - \beta) \xrightarrow{d} \mathcal{N}\left(0, \frac{576}{16}\right) = \mathcal{N}(0, 36).$$

Undoing the $\sqrt{n}$ scaling gives

$$\text{SD}(\hat{b}) \approx \sqrt{\frac{36}{100}} = \frac{6}{\sqrt{100}} = 0.6.$$

This is the SD of the large-sample approximation, not an exact finite-sample formula.

Question 3 (hard). Correct answer: B, (-1; 1).

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 2$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

Convert the correlation into the auxiliary regression slope:

$$\text{Cov}(X, Z) = \frac{1}{2} \cdot 2 \cdot 4 = 4, \quad \pi = \frac{4}{2^2} = 1.$$

Thus $\delta\pi = 3$ and $\beta = b_{\text{short}} - \delta\pi = 2 - 3 = -1 < 0$. But the short OLS slope converges to the positive association:

$$\hat{b}_{\text{short}} \xrightarrow{p} 2 > 0, \quad \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) = 1.$$

More data make the positive association increasingly precise even though the causal effect is negative. Neither a large sample nor sample residual orthogonality repairs the omitted-variable distortion.

Form 4S6B — worked solutions

Question 1 (easy). Correct answer: D, 10.

1. The causal model is

$$Y = \alpha + 11X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

The auxiliary population slope and omitted-variable distortion are

$$\pi = \frac{\text{Cov}(X, Z)}{\text{Var}(X)} = \frac{-2}{6} = -\frac{1}{3}, \quad \delta\pi = 3 \left(-\frac{1}{3} \right) = -1.$$

Hence $b_{\text{short}} = \beta + \delta\pi = 11 - 1 = 10$. The association is one unit below the causal effect. This is a population projection identity, not a formula for exact finite-sample unconditional bias.

Question 2 (medium). Correct answer: C, (4; 0.7).

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 784)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

The LLN gives $D_n \xrightarrow{p} \text{Var}(X) = 4$. Dividing the numerator by 4 divides its variance by $4^2 = 16$:

$$\sqrt{n}(\hat{b} - \beta) \xrightarrow{d} \mathcal{N}\left(0, \frac{784}{16}\right) = \mathcal{N}(0, 49).$$

Undoing the $\sqrt{n}$ scaling gives

$$\text{SD}(\hat{b}) \approx \sqrt{\frac{49}{100}} = \frac{7}{\sqrt{100}} = 0.7.$$

This is the SD of the large-sample approximation, not an exact finite-sample formula.

Question 3 (hard). Correct answer: E, (-0.5; 1).

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 2.5$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

Convert the correlation into the auxiliary regression slope:

$$\text{Cov}(X, Z) = \frac{1}{2} \cdot 2 \cdot 4 = 4, \quad \pi = \frac{4}{2^2} = 1.$$

Thus $\delta\pi = 3$ and $\beta = b_{\text{short}} - \delta\pi = 2.5 - 3 = -0.5 < 0$. But the short OLS slope converges to the positive association:

$$\hat{b}_{\text{short}} \xrightarrow{p} 2.5 > 0, \quad \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) = 1.$$

More data make the positive association increasingly precise even though the causal effect is negative. Neither a large sample nor sample residual orthogonality repairs the omitted-variable distortion.